

WHAT LIQUIFIES A HYDROGEL?

Variable Investigation - Temperature, Acidic Conditions, and Basic Conditions

THE CHALLENGE

Gel-like products such as hand sanitizer can spread and become much thinner when they leave a bottle and reach your hand. In this activity, your team will test ONE possible factor that may change the physical state of a model hydrogel. Your results will be combined with the rest of the class to determine which conditions have the strongest effect.



SCAN FOR QUICK STEPS

How the class is organized

Teams	Variable	Student 1	Student 2	Student 3
Teams 1-4	Temperature	10 C	Room temperature	37 C
Teams 5-7	Acidic pH	pH 5	pH 4	pH 3
Teams 8-10	Basic pH	pH 9	pH 10	pH 11

Safety

- Wear splash goggles throughout the experiment. Wear gloves when directed by your instructor.
- Do not touch or taste acidic or basic test solutions or treated hydrogel.
- Temperature teams: heat only water in a beaker. Never place a plastic Petri dish on a hot plate.
- If a solution spills, a dish cracks, or you are unsure of a step, stop and ask.
- Bulk hydrogel goes in the lined trash container - not down the sink.

Per-team materials

Team type	Materials
All teams	3 x 100 mm x 15 mm Petri dishes; 3 spoons; graduated cylinder(s); 3 pre-portioned 0.25 g Water Gel samples; DI water; timer; goggles
Temperature	Thermometer; access to shared cold-water bath; beaker + hot plate for warm-water condition
Acidic pH	Pre-prepared and labeled pH 5, pH 4, and pH 3 solutions; pH paper/indicator as directed
Basic pH	Pre-prepared and labeled pH 9, pH 10, and pH 11 solutions; pH paper/indicator as directed

STUDENT PROCEDURE

Each student runs one condition. Record results on the separate data sheet.

FAIR TEST RULE

Only your assigned temperature or pH should change. Hydrogel amount, water volume, stirring, and waiting time must stay the same for all three students.

Common sample procedure - everyone does this

- 1 Label your three Petri dishes with your team conditions.
- 2 Place one pre-portioned 0.25 g Water Gel sample into each Petri dish.
- 3 Add 20 mL DI water to each dish while stirring gently with a spoon.
- 4 Allow each hydrogel sample to hydrate for 2 minutes.
- 5 Prepare each assigned test liquid. Measure and record its actual temperature or pH immediately before addition.
- 6 Add 20 mL of the assigned test liquid to the matching hydrogel sample.
- 7 Stir each sample exactly 10 strokes using the same motion.
- 8 Wait 5 minutes.
- 9 Score the hydrogel from 1 to 5, note whether free water is visible, and write one short observation.

Variable-specific instructions

TEMPERATURE TEAMS

Cold: chill DI water in a refrigerator; let it warm to about 10 C, verify, then measure 20 mL.

Room: use 20 mL DI water at room temperature.

Warm: heat DI water in a beaker to about 37 C; verify, then measure 20 mL. Record temperature before addition.

ACID TEAMS

Student 1: 20 mL pH 5 solution.
Student 2: 20 mL pH 4 solution.
Student 3: 20 mL pH 3 solution.
Confirm/record the actual pH as directed before adding the solution.

BASE TEAMS

Student 1: 20 mL pH 9 solution.
Student 2: 20 mL pH 10 solution.
Student 3: 20 mL pH 11 solution.
Confirm/record the actual pH as directed before adding the solution.

Hydrogel rating scale

Score	Description	What it looks like
5	Fully solid gel	No obvious change
4	Slightly smaller / softened	Small change
3	Partially liquified	About half gel-like / half liquid
2	Mostly liquid	Only a small amount of gel structure remains
1	Fully liquid	Gel structure is no longer apparent

OBSERVATION TIP

Also record whether you can see free water released from the gel. Two samples can receive the same 1-5 score but still look different.

HYDROGEL VARIABLE TEST

Data Sheet

ONE TEAM / ONE VARIABLE / THREE CONDITIONS

STUDENT NAMES _____

ASSIGNED VARIABLE

☐

Temperature

☐

Acidic pH

☐

Basic pH

TEMPERATURE

10 C cold | room temp | 37 C warm

ACIDIC pH

pH 5 | pH 4 | pH 3

BASIC pH

pH 9 | pH 10 | pH 11

Same procedure for every sample

- 1 Add 0.25 g hydrogel powder + 20 mL DI water to your Petri dish.
- 2 Stir gently and allow the gel to hydrate for 2 minutes.
- 3 Measure and record the temperature or pH of the test liquid, then add 20 mL to the gel.
- 4 Stir exactly 10 strokes. Wait 5 minutes.
- 5 Score the gel, mark whether free water is visible, and write observations.

Hydrogel score

5 solid / no change**4** softened**3** partly liquefied**2** mostly liquid**1** fully liquid

Record your three samples

Sample	Target	Actual temp / pH	Score 1-5	Free water?	Observations
1			_____	YES / NO	
2			_____	YES / NO	
3			_____	YES / NO	

Quick conclusion

1. Which of your three conditions changed the gel the most?

2. What evidence supports your answer?

3. Would your variable alone explain why a sanitizer-like gel changes when it reaches a hand? Why or why not?

CLEANUP

Put used hydrogel in the lined trash container - do not put bulk gel down the sink.

Rinse reusable glassware/spoons only after the bulk gel has been removed.